KIDASHI*sim*

USER MANUAL

A Guide to the KidashiSim Showcase & Its Agent-Based Model

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OPEN-SOURCE RESEARCH SOFTWARE
 · github.com/Adedeji-Taiwo/kidashi_abm
Live app: kidashiabm.streamlit.app

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BUILT WITH

MESA FRAMEWORK · SALIB · UM6P · XCHANGEBOX SOLUTIONS · STREAMLIT

Independent academic research software developed in connection with XchangeBox Solutions Ltd. operational data (anonymised) and UM6P coursework. Figures, prices and simulation output shown throughout this manual are illustrative.

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SECTION 1

Getting Started — Navigation

KidashiSim uses a simple, shareable-URL navigation model: a hero landing view, a Case Studies grid, an interactive Simulator, a Documentation view, and an About dialog. The model can be explored two ways — interactively in the Simulator (Section 4), directly in the browser, or in full via the KidashiABM codebase (Section 7).

1.1 The Home Screen

The landing view opens on FIG 01 (see Section 2) beside a short project summary and three actions: **Explore case studies**, **Read documentation**, and **About**.

1.2 Case Studies

A grid of market-context cards, each tagged LIVE or SOON. The live Nigeria card's **Launch simulator** button opens the in-app Simulator (Section 4) directly, without leaving the page; SOON cards are illustrative roadmap placeholders. See Section 3.

1.3 The Simulator

Reached from the live Nigeria card, the Simulator runs KidashiModel directly in the browser: set a scenario with eight controls, click **Run simulation**, and read back six headline metrics plus three time-series charts. See Section 4.

1.4 Documentation

An in-app summary of the model — overview, agents, shock regimes and diversity policies — condensed from the full technical reference in Sections 7–8 of this manual.

1.5 About

A modal dialog with author information, licensing, and the citation to use when referencing this software. See Section 6.

> **TIP** Case Studies, Documentation and the Simulator are reachable by URL (`?view=cases`, `?view=docs`, `?view=simulator`), so any of the three can be bookmarked or linked to directly.

SECTION 2

Reading FIG 01 — The Live Figure

FIG 01 is the centrepiece of KidashiSim: a live canvas animation showing how farmers, Kidashi credit, the Kano market, downstream traders and export demand interact under a chosen shock regime. Everything in the figure is generated client-side and re-computed on every animation frame — the numbers in its header are read live from the agents currently on screen, not hard-coded.

2.1 Nodes & Routes

Node	What it represents
Farm Clusters	Where maize, sorghum and tomato are grown; home position for farmer agents and Kidashi trust-circles.
Kidashi Credit	The trust-circle liquidity source; sends periodic credit pulses outward.
Kano Market	Where farmgate prices form; hosts the three reactive market crates.
Lagos / Processors	Downstream aggregation and clearing for produce leaving the market.
Export Link	The international trade channel; the origin point of trade-shock ripples.

2.2 Agent States

State	Colour	Meaning
Stable	Green	Diversified or otherwise unaffected by the current shock.
Credit	Cyan, ringed	Has active Kidashi credit; more resilient to distress sales.
Vulnerable	Gold	Exposed to the active shock but not yet selling at a loss.
Distress	Red, trailed	Selling under pressure; leaves a short fading trail as it moves.

2.3 Trust Circles & Credit Pulses

Six small trust-circles orbit the farm cluster, each representing a Kidashi credit group. A cyan pulse periodically travels from the Kidashi Credit node to one circle, brightens it, and has a chance to upgrade one member's state (distress -> vulnerable -> credit) — a direct visualisation of the trust-circle credit mechanic described in Section 7.3.

2.4 Reactive Market Crates

Three small crates beside the Kano Market node — maize, sorghum, tomato — fill and flash on every farmgate transaction, giving the market a visible pulse of activity in the same way the shelf grid does in related retail-side ABM work, while representing KidashiABM's own crop-lot bookkeeping.

2.5 Shock Regimes

The regime shown in FIG 01 matches KidashiModel's `--shock` flag directly, and drives a distinct visual effect:

Regime	Weather / Trade	Visual effect in FIG 01
baseline	Low / Low	No overlay — calm conditions.
climate_stress	High / Low	Diagonal amber rain-streaks drift across the canvas.
trade_disruption	Low / High	Red-amber ripples expand from the Export Link node.
compound	High / High	Both effects together.

> **TIP** FIG 01 respects `prefers-reduced-motion`: with that setting on, it renders a single still frame instead of animating.

SECTION 3

Case Studies Reference

Four market-context cards are presented on the Case Studies view. Only the Nigeria card is backed by a live model at the time of writing — its **Launch simulator** button opens the Simulator (Section 4); the remaining three are illustrative roadmap placeholders and should be replaced with the project's actual pipeline before publishing.

Market	Status	Focus
Nigeria	*LIVE*	Kidashi trust-circle credit, crop diversification and farmgate price resilience for maize, sorghum and tomato across Kano and Jigawa.
Ghana	*SOON (placeholder)*	Liquidity access, aggregator financing and price pass-through for smallholder cocoa.
Kenya	*SOON (placeholder)*	Cold-chain financing and export-channel liquidity for horticulture cooperatives.
Senegal	*SOON (placeholder)*	Trust-circle credit uptake and distress-sale pressure for groundnut cooperatives.

SECTION 4

Simulator Reference

The Simulator runs KidashiModel directly in the browser — the same model the CLI and experiment runners use — and returns headline metrics and time-series charts without leaving the page.

Accessing the Simulator

From the Case Studies view, the live Nigeria card's **Launch simulator** button opens the Simulator in place — the same tab, no new window — by setting the app's internal view state and re-rendering, exactly like the Case Studies, Documentation and About buttons. The URL `?view=simulator` opens it directly.

Scenario Controls

Eight controls mirror `run_model.py`'s command-line arguments exactly (Section 7.1), so a scenario configured in the Simulator can be reproduced identically from the CLI with the same flags and seed:

Control	Range	Default	CLI equivalent
Simulation steps	10 – 80 seasons	40	--steps
Farmer agents	50 – 500	100	--num_farmers
Trader agents	5 – 20	10	--num_traders
Fintech penetration	0 – 100%	40%	--fintech_penetration
Trade openness	0 – 100%	20%	--trade_openness
Diversity policy	none / incentive / mandate	none	--diversity_policy
Shock regime	baseline / climate_stress / trade_disruption / compound	baseline	--shock_regime
Random seed	any integer	42	--seed

Clicking **Run simulation** instantiates KidashiModel with the chosen parameters and steps it forward. Identical parameter combinations are cached, so re-running the same scenario a second time is instant rather than re-simulating from scratch.

Reading the Results

A results panel appears below the controls once a run completes, in two layers:

Layer	Contents
Metric cards	Mean farmer wealth, maize / sorghum / tomato price, diversity index, and PAR30 (credit risk) — all read from the final simulated season.
Charts	Farmgate prices over time (maize, sorghum, tomato), mean farmer wealth over time, and diversity index over time — each plotted against simulation step, drawn from the same Mesa DataCollector Section 7.6 describes.

TIP Every chart and metric reflects only the scenario most recently run. Adjust the controls and click **Run simulation** again to compare a different configuration — the parameters behind the current results are shown as a caption line above the metric cards.

SECTION 5

Documentation Page Reference

The in-app Documentation view condenses the material in Sections 7–8 of this manual into four short blocks.

Overview

An Agent-Based Model investigating how fintech-led liquidity provision (the Kidashi trust-circle credit product) and production diversity jointly shape farmgate price resilience and smallholder welfare under domestic supply shocks and international market disruption.

Headline figure	Meaning
55% spoilage reduction	Post-harvest tomato spoilage reduction with fintech (TomatoPro-validated).
18.6% profit uplift	Producer profit uplift under financial liquidity.
Under 72 hours	Kidashi payment cycle, versus a 60–90 day baseline without fintech.

Agents, shock regimes & diversity policies

Reproduced in full in Sections 7.2–7.4 (agents), 2.5 (shock regimes) and 7.6 (diversity policies) of this manual.

SECTION 6

About & Citation

This application showcases an academic Agent-Based Model exploring how fintech-led liquidity and crop-diversification policy shape farmgate price resilience and smallholder welfare in Nigerian staple value chains. All figures shown in the live app are illustrative simulation output.

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Citation

> Adedeji, Taiwo M. (2026). *KidashiABM — Agent-Based Modelling of Fintech-Led Liquidity Provision and Farmgate Price Resilience in Nigerian Staple Value Chains.* Source: github.com/Adedeji-Taiwo/kidashi_abm · Live app: kidashiabm.streamlit.app

License

SECTION 7

Model Architecture & Technical Notes

This section documents KidashiModel itself, independent of the KidashiSim showcase site.

7.1 Computational Framework

Built on Mesa (Python ABM framework). Each model step advances the simulated season, updating agent decisions, farmgate price formation, and any active shock. Runs are fully seeded: identical seeds produce bit-identical results.

7.2 Farmer Agents (100–1000 per run)

Adaptive mean-variance portfolio allocation across maize, sorghum and tomato; stochastic production with weather-correlated shocks (a shared z-score, correlation 0.60, plus idiosyncratic noise, replicating the spatial structure of CHIRPS/ERA5-Land rainfall fields); multi-channel sales with a distress-discount mechanism.

7.3 FintechProvider Agent (1 per run)

Trust-circle credit scoring; Kidashi / Farm-to-Factory disbursement logic; Portfolio-at-Risk (30-day) tracking.

7.4 Trader Agents (5–20 per run)

Aggregator/processor procurement; two-scenario spoilage modelling; export-channel price transmission.

7.5 Market Mechanism

Farmgate prices form endogenously via an inverse supply function, modulated by: aggregate production across all farmer agents; an exogenous trade-shock multiplier (Europe-linked demand disruption); and fintech-enabled reduction in distress-sale discounts.

7.6 Key Model Features

Feature	Role
Shannon Diversity Index (H')	Computed per farmer each step; used as both a policy lever and a credit-scoring signal.
Gini coefficient	Collected via Mesa DataCollector for distributional analysis of farmer wealth.
Nutrition proxy score	Caloric adequacy derived from farm income relative to a 5-person household subsistence threshold (WFP 2023).
Correlated weather shocks	Shared z-score (correlation 0.60) across farmers plus idiosyncratic noise.
Reproducibility	All runs seeded; identical seeds produce bit-identical results.

Diversity policies — none: no intervention, portfolio shaped by expected utility alone. **incentive:** diversified farmers (H' above threshold) receive a 5% price premium. **mandate:** minimum crop-share floor raised to equal weighting across all three crops.

> TIP Diversity policy and shock regime can be combined in the CLI (`--policy incentive --shock compound`) to trace how the two levers interact — see the project README's Experiment Scenarios table.

SECTION 8

Calibration & Data Sources

Parameters representing XchangeBox operational data are anonymised and used under professional confidentiality; only aggregate or published values are reported here.

Parameter	Value	Source
Maize farmgate price baseline	NGN 310,000/MT	CBN / NBS Commodity Price Data (2023)
Tomato farmgate price baseline	NGN 420,000/MT	CBN / NBS (2023)
Maize yield range	1.2–3.5 MT/ha	IITA on-farm trial data (BASICS-II)
Tomato spoilage, no fintech	35%	TomatoPro baseline (XchangeBox, 2025)
Spoilage reduction, with fintech	55%	TomatoPro validated result
Credit yield uplift	12%	Estimated from 18.6% profit uplift, TomatoPro
Payment cycle, no fintech	75 days	XchangeBox field data (Kano/Jigawa, 2025)
Payment cycle, with fintech	Under 72 hours	Kidashi product
Farm size distribution	Log-normal (mean 0.7, sd 0.5 ha)	NBS Agricultural Survey (2021)
Weather–yield correlation	0.60	ERA5-Land / CHIRPS
Subsistence threshold	NGN 180,000/season	WFP Northern Nigeria (2023)

SECTION 9

Glossary & Citation

9.1 Glossary of Key Terms

ABM — Agent-Based Model — a simulation in which individual agents follow rules and interact to produce emergent system behaviour.

Distress Sale — A below-market sale a farmer makes under acute liquidity pressure.

Fintech Penetration — The share of farmers with active Kidashi credit in a given run.

Gini Coefficient — Inequality measure applied to farmer wealth; 0 = perfect equality, 1 = maximum inequality.

Kidashi — The trust-circle credit product at the centre of this model; disburses liquidity through small farmer groups.

PAR30 — Portfolio-at-Risk over 30 days — a credit-quality metric tracked by the FintechProvider agent.

Shannon Diversity Index (H') — A measure of how evenly a farmer's land is split across maize, sorghum and tomato.

Shock Regime — The combination of weather volatility and trade-shock probability active in a run: baseline, climate_stress, trade_disruption or compound.

Simulator — The in-app scenario runner (Section 4) that instantiates KidashiModel directly in the browser and returns headline metrics and time-series charts for the chosen parameters.

Trust Circle — A small group of farmers who jointly qualify for and receive Kidashi credit.

9.2 Citation

How to Cite This Software

> Adedeji, Taiwo M. (2026). *KidashiABM — Agent-Based Modelling of Fintech-Led Liquidity Provision and Farmgate Price Resilience in Nigerian Staple Value Chains.*
> Software available at: github.com/Adedeji-Taiwo/kidashi_abm · Live app: kidashiabm.streamlit.app